

Chapter 110 — Overactive Bladder — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **OAB (overactive bladder)** = urgency $\pm$ frequency, nocturia, UUI (urgency urinary incontinence); no UTI (urinary tract infection) or obvious pathology.
- **OAB = symptom; DO (detrusor overactivity) = urodynamic finding** — not interchangeable.
- **Minimum evaluation:** history + examination + urinalysis.
- **No urodynamics, cystoscopy or ultrasound** in uncomplicated initial work-up.
- **Chronic antimuscarinic >3 months** $\rightarrow$ **dementia risk $\uparrow$ 10–65%.**
- **Preferred drug:** β 3 agonist, or ER trospium / darifenacin / fesoterodine.
- **DO in men with UUI >90%;** absent in 30–40% of women with UUI.
- **PVR (postvoid residual) +10 mL $\rightarrow$ catheterisation +27%** after onabotulinumtoxinA.
- **SNS (sacral nerve stimulation):** 70–80% success at 5 y; revision >30%.
- **AUA/SUFU (American Urological Association / Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction) lines $\neq$ mandatory sequence;** no treatment is acceptable.

TERMINOLOGY AND DEFINITIONS

- **Four symptoms:** daytime frequency, nocturia, urgency, UUI.
- **Normal daytime frequency ≤ 7 ;** bother from ≥ 8 voids (every ≤ 2 h).
- **Nocturia ≥ 1** by definition; **bother ≥ 2 ,** marked ≥ 3 .
- **Nocturia:** each void preceded *and* followed by sleep.
- **Urgency:** sudden, compelling desire, difficult to defer.
- **Most bothersome LUTS (lower urinary tract symptom), both sexes:** UUI.
- **MUI (mixed urinary incontinence)** = SUI (stress urinary incontinence) + UUI ("OAB wet").
- **Isolated nocturia $\rightarrow$** think nocturnal polyuria or poor sleep.
- **OAB vs BPS (bladder pain syndrome):** fear of leakage vs fear of pain.
- **BPS filling:** steady accelerating pain; **OAB:** phasic urgency waves.
- **BPS:** consistent volumes, can defer; suprapubic pain; no UUI.

EPIDEMIOLOGY AND ECONOMICS

- **Prevalence:** OAB $\approx$ 12–29%; UUI $\approx$ 1.5–17%.
- **Women vs men:** OAB 1.5–2 $\times$; UUI 2–3 $\times$.
- **Highest:** EpiLUTS (loose definition); **lowest:** EPIC (ICS, International Continence Society, definition).
- **NOBLE:** ≥ 4 urgency episodes/4 wk + ≥ 8 voids/day or coping.
- **NOBLE:** OAB similar by sex; **UUI $\approx 3\times$ in women.**
- **Age rise:** women 40s, men 50s–60s; UUI after 44 (F) / 64 (M).
- **Slightly highest by race:** Black men and women.
- **Incidence** 3–8%/y (women); **progression to UUI $\approx 1\%/y$;** remission $\approx 6\%/y$.
- **<6% take medication;** $\approx 92\%$ fail anticholinergic by 2 y.
- **Half never refill** first prescription; most stop by 6 months.
- **Stop reasons:** unmet expectations > side effects > cost.
- **Most cost-effective third-line:** onabotulinumtoxinA 100 U; costliest: SNS.

PATHOPHYSIOLOGY AND ETIOLOGY

- **DO in patients with urgency:** 44–69%; urgency in DO $\approx$ half.
- **No animal model of OAB** — urgency cannot be reported.
- **Main regulatory level:** periaqueductal grey + PMC (pontine micturition centre).
- **A δ fibres** = normal volume sensing; **C fibres** = "silent", recruited in disease.
- **Two bladder-centric theories:** urothelium-based; myogenic (patchy denervation, micromotions).
- **Micromotions coordinated** across bladder $\rightarrow$ DO.

- **Bowel crosstalk:** rectal distension alters capacity, sensation, DO.
- **BOO (bladder outlet obstruction):** indirect, age-accelerating; not direct cause.
- **Anxiety $\rightarrow$** heat-pain temporal summation $\uparrow$ (central sensitisation).

CLINICAL ASSESSMENT

- **Faecal incontinence:** 7–15% general vs $\approx 50\%$ of women with OAB.
- **Antimuscarinic relative CI:** closed-angle glaucoma (uncontrolled), cognitive impairment, retention, impaired gastric emptying.
- **β 3 agonist relative CI:** uncontrolled hypertension, arrhythmia.
- **Most important examination:** pelvic/genital $\pm$ DRE (digital rectal examination).
- **Women:** check prolapse, atrophic vaginitis, caruncle, labial adhesions.
- **All:** perineal sensation, anal tone, cognition (MMSE, Mini-Mental State Examination), frailty (TUGT, Timed Up and Go Test, 10 ft).
- **PVR not needed** before first/second line in uncomplicated patients.
- **PVR needed:** men with voiding LUTS, retention risk, prior outlet/sling surgery, neurological disease.
- **PVR method:** bladder scanner first; catheter if unavailable.
- **Urinalysis mandatory;** culture only if suspected; creatinine and PSA (prostate-specific antigen) not routine.
- **Diary:** 3–7 days ideal; minimum 24–48 h.
- **Max voided volume** = functional capacity.
- **Frequency count** includes last void before sleep, first on rising.
- **Urgency scale:** 0 social $\cdot$ 1 normal $\cdot$ 2 passed $\cdot$ 3 no leak $\cdot$ 4 leak.
- **AUA PROMs (patient-reported outcome measures):** UDI (Urogenital Distress Inventory) / UDI-6, IIQ (Incontinence Impact Questionnaire), OAB-q.
- **OAB-V8** = screening; **QUID** = incontinence type; **PRAFA** = treatment effect.
- **LURN SI** = both sexes, 29 or 10 items.
- **Urodynamics if:** changes invasive/irreversible decision; multiple failures; BOO after outlet surgery.
- **Not if:** index patient (woman, low PVR, no prolapse, no neuro, minimal SUI).
- **OAB dry DO:** men 60–70%, women 40–50%.
- **DO correlation:** best UUI, worst nocturia.
- **No DO $\rightarrow$** usually increased bladder sensation.

- **Fast filling** induces DO; **supine second fill** reduces DO to unmask SUI.
- **DO persists after TURP (transurethral resection of prostate)** in up to 60%.

MANAGEMENT

- **First line:** education + behavioural; placebo response large.
- **Fluid ↓ 25%** meaningfully reduces symptoms.
- **Caffeine ↓** : helps urgency/frequency, not reliably UUI.
- **Bariatric surgery:** each 5% weight loss → 7–8% ↓ incontinence risk.
- **PFMT (pelvic floor muscle therapy)/bladder training:** 50–80% fewer incontinence episodes.
- **Antimuscarinic vs placebo:** cure likelihood ↑ 20–70%; high withdrawal.
- **Extended-release (ER) over immediate-release** antimuscarinics.
- **Men, voiding LUTS:** α1-blocker first-line; add antimuscarinic/β3 for storage.
- **Antimuscarinic ± α1-blocker:** retention up to 3% → caution if PVR raised.
- **Minimum trial:** 4–12 weeks (behavioural and drug).
- **Third line:** PTNS (percutaneous tibial nerve stimulation), onabotulinumtoxinA, SNS.
- **PTNS:** 60–80% response; weekly ×12 wk; response wk 8–10.
- **OnabotulinumtoxinA:** UUI ↓ 59%; ≈25% dry; UTI ≈20%.
- **OnabotulinumtoxinA CISC (clean intermittent self-catheterisation):** 0–12% trials; ≈20% women, ≤35% men real world.

- **SNS:** all eventually need battery change.
- **Frailty:** 1-year morbidity up to 50% after minor procedures.
- **Operate (augmentation/diversion) only** if refractory to all; weigh vs pads/catheter.
- **Aggressive treatment justified:** incontinence-associated dermatitis.

CLASSIC TRAPS

- **Trap:** SUI + OAB dry = MUI — **Actually:** MUI needs UUI.
- **Trap:** urinary + faecal = mixed — **Actually:** double incontinence.
- **Trap:** OAB requires DO — **Actually:** symptom diagnosis; DO often absent.
- **Trap:** UUI in women predicts DO — **Actually:** absent in 30–40%.
- **Trap:** BOO causes OAB — **Actually:** indirect; DO persists after TURP.
- **Trap:** urodynamics before first treatment — **Actually:** not in uncomplicated.
- **Trap:** PVR before oral drugs always — **Actually:** only if complicated.
- **Trap:** lines are sequential — **Actually:** combine or skip per goals.
- **Trap:** onabotulinumtoxinA retention 0–12% — **Actually:** real world 20–35%.
- **Trap:** C fibres large-calibre — **Actually:** smallest, unmyelinated.